$\left[\hat{\sigma}_x^{(i)}, \hat{\sigma}_z^{(j)}\right] = 0$ for $i \neq j$, the exponential $\mathrm{e}^{\mathrm{i}H_{\mathcal{E}}\tau}$ factors into single-spin terms,

$$\mathrm{e}^{\mathrm{i}\hat{H}_{\mathcal{E}}\tau} \equiv \mathrm{e}^{\mathrm{i}\sum_i \hat{H}_{\mathcal{E}}^{(i)}\tau} = \prod_i \mathrm{e}^{\mathrm{i}\hat{H}_{\mathcal{E}}^{(i)}\tau}. \tag{5.171}$$

Therefore we can write the environment self-correlation function (5.170) as

$$\mathcal{C}(\tau) = \sum_{ij} g_i g_j \left\langle \mathrm{e}^{\mathrm{i}\hat{H}_{\mathcal{E}}^{(i)}\tau} \hat{\sigma}_z^{(i)} \mathrm{e}^{-\mathrm{i}\hat{H}_{\mathcal{E}}^{(i)}\tau} \hat{\sigma}_z^{(j)} \right\rangle_{\hat{\rho}_{\mathcal{E}}} \equiv \sum_{ij} g_i g_j \left\langle \hat{\sigma}_z^{(i)}(\tau) \hat{\sigma}_z^{(j)} \right\rangle_{\hat{\rho}_{\mathcal{E}}}. \tag{5.172}$$